Relationship Between the Street Spatial Environment and Crime Based on Quantitative Network Analysis: A Case Study in Guangzhou, China

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Abstract

Reduction in criminal activity would improve the quality of life and social well-being of people. Street space, as a basic element of the cognitive environment of criminals, influences the spatial distribution of criminal activities. In existing studies, the construction and calculation of street networks are too ideal, the differentiation of different pedestrian flows lacks precision, and the selection of indicators of street space characteristics lacks richness. This scenario leads to vague research results and creates difficulty in effectively guiding the related urban management. Therefore, by constructing a complex network of urban street space and using pedestrian flow data to characterize the passage potential and surveillance level of different pedestrian flows on each street, this study performed a negative binomial regression analysis to compare criminal activities with the street features of built environment, street users, and social structure. Streets with higher potential for traveler passage are found to be more anonymous and prone to crime, whereas the opposite is true for streets with high potential for local passage. Meanwhile, the distribution of various types of public facilities in the street and the mobility of the street will have an impact on criminal activities. Results of this study can help in proposing appropriate planning and policy recommendations for the refined management of urban to maintain a stable life for residents.

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CCS Concepts

• **Social and professional topics** → User characteristics; Geographic characteristics.

Keywords

street, crime, spatial environment, quantitative network analysis, city management

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1 Introduction

As the process of urbanization accelerates and the functions and size of cities become increasingly larger and more complex, crime has become an important aspect of daily security in cities, posing a risk to people's quality of life and the well-being of society [1, 2]. According to crime pattern theory, criminals form cognitive spaces

through their daily activities and seek suitable opportunities to commit crimes from them. Among them, street space, as a basic element of criminals' cognitive space, plays an important role in the formation of crime space distribution [3]. In the current research related to public safety management, the relationship between urban street space and crime has lacked adequate research, and no systematic theory has been formed. For example, Kim et al. introduced the concept of betweenness centrality of street networks to measure the accessibility of each street and found that under the combined effect of crime escape convenience and pedestrian flow monitoring, the crime rate shows a tendency of increasing and then decreasing with the increase in street accessibility [4]. Davies et al. found that more linear streets were less likely to experience criminal activity than meandering streets, confirming the strong relationship between street network attributes and burglary [5]. However, this measure of accessibility is relatively simplistic based only on calculating the betweenness centrality of a roadway and lacks the subjective influence of factors such as pedestrian flow and activity on street access choices.

Street pedestrian flow as a complex factor, which can bring frequent flow of property and goods to increase the target of criminals' actions and improve criminal activities and as an important part of the informal social control of street space, has an impact on the control of criminal activities [6, 7]. According to social disorganization theory, several mobile and heterogeneous street pedestrian flows will have a greater impact on criminal activity, which is generally reflected in the different types of flows between locals and non-locals [8]; however, the existing studies are relatively homogeneous in the way they differentiate between locals and non-locals in the streets. For example, Frith et al. modeled the movement of local versus non-local passersby by limiting the computational radius of street betweenness centrality to varying centrality and found that non-local foot traffic was associated with an increase in criminal activity [9]. Kim et al. weighted the betweenness centrality of streets by the number of residential properties, residential population, and employees in businesses, each representing a different type of foot traffic for street crime analysis [4]. Apart from the large relationship between different types of pedestrian flow on the generation of criminal activities on streets, attention should also be paid to other spatial street characteristics. For example, Hu and Wang used spatial syntax to study the relationship between integration and connectivity of street space and criminal activities based on the CPTED and found that street space with high integration and connectivity facilitates the formation of natural surveillance, which reduces criminal activities [10]. In the existing research on street crime, the construction and calculation of the street network is found to be too ideal and simple. In addition, the distinction between different types of pedestrian flow lacks precision, and the types of indicators to explore the relationship between the street space characteristics and criminal activities lack richness.

To address the above challenges, this study utilizes mobile signaling data as well as traveler's trajectory record data and creates a refined distinction of different types of pedestrian flow on the street, including local permanent residence, local floating residence, and travel. The mobile signaling data have the advantages of a large sample size, timeliness, and spatial and temporal coverage; it can also promptly and precisely characterize the distribution pattern of the population and mobility [11]. The refined differentiation of pedestrian flow as the weight of calculating the betweenness centrality of street can effectively avoid the problem of oversimplifying the calculation of road accessibility. In addition, this study makes a regression analysis between the factors characterizing urban street space and criminal activities to obtain detailed conclusions regarding the relationship between urban street space and criminal activities. The results of this study can help in proposing appropriate planning and policy recommendations for the refined management of urban security and maintaining a stable life for the residents.

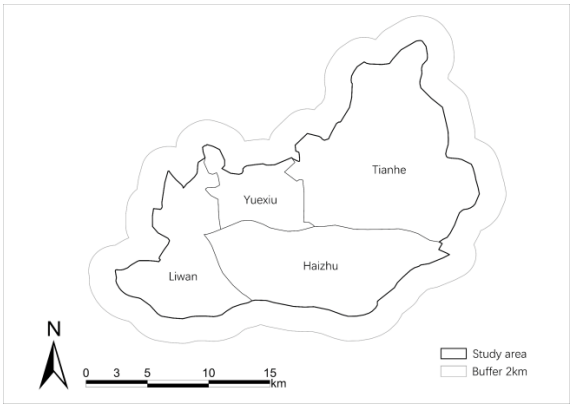

Figure 1: Study Area and Buffer Zones.

2 Study area and data

2.1 Study area

The research area of this study is the central city of Guangzhou, as shown in Figure 1. As a megacity in South China with developed economy, convenient transportation, and a large foreign population, Guangzhou is a hot case city in Chinese crime geography research. Its central urban area includes Tianhe District, Yuexiu District, Liwan District, and Haizhu District. This area is representative in crime research because of the complex-built environment, concentrated population distribution, and outstanding crime quantity.

2.2 Data and indicator system construction

2.2.1 Street Data and Crime Data. This study constructed a street network dataset using the open-source data called open street map. The street data were first filtered to retain the four categories of roads that are closely related to crowd activities, namely expressways, arterial roads, secondary trunk roads, and branch roads. Then, the organized street data were converted into a form suitable for complex network analysis by extracting street intersections as network nodes and street segments between the intersections as connections between the nodes. Street segments close to the study area boundary may inherently have less movement if the street network data are limited to the study area boundary. To avoid this problem, we included street networks within 2 km of the study area boundary.

The crime data come from the 2019–2021 criminal case judgments of the basic courts made public by China Judgments Online (https://wenshu.court.gov.cn/), from which the judgments of crimes occurring on the street level of the four central districts of Guangzhou are screened. The location descriptions are then converted to WGS-84 geographic coordinates, which are aggregated to the nearest street segments using geographic proximity, constituting a crime spatial information dataset, to obtain the number of street crimes in each street segment.

2.2.2 Factors influencing crime. Street space is a complex organic unity. To explore the relationship between urban street space characteristics and criminal activities with comprehensiveness and integrity, this study introduces three dimensions of crime influencing factors in the built environment, street users, and social structure. Table 1 shows its proxy indicators and data sources.

In terms of built environment factors, the type of facilities in a street is an important factor that affects the number and type of people who are familiar with the place, which in turn has an impact on criminal activity [12]. Crime pattern theory categorizes places and facilities with high crime occurrence into concepts such as crime generator and crime attractor [13]. Facilities such as shopping centers, schools, hotels, and so on, are usually regarded as crime generators, which promote the increase in potential criminals and potential victims by attracting a large number of people to the same place. Meanwhile, facilities such as bars, Internet cafes, KTVs, and so on, are usually regarded as crime attractors, which are attractive to motivated criminals because of their specific functions and the characteristics of weak surveillance. Combining relevant theories and urban land use classification, this study selects ten types of facility densities, namely, commercial, business, recreational, cultural, educational and research, medical and healthcare, residential and residential service, parks, industrial, and transportation, as built environment characteristic factors.

In terms of street user factors, this study simulates the intensity of crowd activity and surveillance in street segments using the betweenness centrality of streets weighted by pedestrian flow. To investigate whether different types of crowd activity and surveillance have differential effects on crime, we categorize street flows into local permanent resident, local floating resident, and traveler flows, and measure the population densities of the three types in each street segment. The data of local permanent and floating resident flows are obtained from mobile signaling data, and the data of traveler flows are obtained from the website of "Two Steps" (https://www.2bulu.com/). "Two Steps" is a web platform that serves travelers and can realize the functions of recording and sharing the trajectories of recreational routes, which are data information voluntarily and openly uploaded by users and can objectively reflect the behavioral activities of travelers and the intensity of travelers' flow in different areas.

In terms of social structure factors, social disorganization theory emphasizes the impact of socio-demographic contexts on facilitating or inhibiting the occurrence of crime; it argues a strong correlation between poverty, social deprivation, and high levels of crime; it also suggests that lower economic and social status, greater ethnic heterogeneity, and higher population mobility can weaken formal and informal social control and create a degree of social disorganization, thus fostering the development of crime [8]. In this study, the percentage of immigrant and youth population were selected as social structure factors to investigate the impact of socio-demographic characteristics on crime.

3 Methods

3.1 Betweenness Centrality

Road network has the model properties of complex network with the structural stability of the network, the evolutionary dynamics mechanism of the network, and so on. This study measures the characteristics of street segments of the road network, discovers the key street segments of the road network, and examines the community structure of the network. Accordingly, we choose the betweenness centrality index in social network analysis. Betweenness centrality describes the proportion of all shortest paths in the road network that passes through a certain street segment. If the betweenness centrality of a street segment is extremely high, it reflects that this street segment is the necessary route for the shortest paths of many street segments. This finding indicates that it assumes an important transit role in the road network. Betweenness centrality can well describe the importance of the corresponding street segment in the network, and its calculation method is:

$$B_e = \sum_{i,j \in V, i \sim j} \frac{\sigma_{ij}(e)}{\sigma_{ij}}, \tag{1}$$

where σ_{ij} denotes the number of shortest paths between node i, and $\sigma_{ij}(e)$ denotes the number of paths which pass through the segment e.

This study classifies the flow of people into local permanent resident flow, local floating resident flow, and traveler flows, according to the different natures of the flow of people weighted for the calculation of betweenness centrality [4]. According to the start and end street segments of the flow of people to the betweenness centrality of the weighted assignment of value, the specific formula is as follows:

$$B_e = \sum_{i,j \in V, i \sim j} \frac{\sigma_{ij}(e)\, w_i w_j}{\sigma_{ij} \sum w_k}, \tag{2}$$

where w_i and w_j denote the weights of the start and end segments, respectively. $\sum w_k$ is the weight of all possible destinations.

The betweenness centrality constructed by weighting the flow edges can indicate both the different uses of the same street segment by different flows as well as the magnitude of the pedestrian passage potential and the amount of pedestrian surveillance, which facilitates in-depth exploration of the relationship between different natures of flows and crime.

3.2 Negative Binomial Regression

The phenomenon of crime is unevenly distributed in space with some aggregated areas, statistically presenting the phenomenon of over-discrete distribution, that is, the variance is significantly larger than the mean; thus, instead of the traditional Poisson regression model [17], the negative binomial regression model is used for analysis [18].

Negative binomial is a special type of regression model for describing some negative effects and is suitable for fitting a non-linear

Table 1: Elements and Proxy Indicators of Influence on Crime.

Form	Indicator	Explanation and Meaning		Sources
Features of the built environment	Density of commercial facilities	Sum of the number of small stores, restaurants, bars, newsstands, convenience stores, supermarkets, large shopping malls, hotels, auto repair stores, building materials stores, furniture stores, marketplaces, and vegetable markets	Count POI data points within 55 m on either side of the centerline of each street segment as facility points within the street frontage space of the segment [14]	Amap POI data
	Density of business facilities	Sum of number of banks, companies		
	Density of recreational facilities	Sum of the number of KTVs, Internet cafes, clubs, nightclubs, dance halls, chess and card rooms, amusement halls, foot bath centers, bathing centers, and massage parlors		
	Density of cultural facilities	Sum of the number of cultural activity centers, museums, and art galleries		
	Density of educational and scientific research facilities	Sum of the number of elementary school, secondary schools, vocational and technical schools, specialized schools, colleges and universities, research institutes		
	Density of medical and health facilities	Sum of number of hospitals, clinics		
	Density of residential and residential service facilities	Sum of the number of residential neighborhoods and kindergartens		
	Density of Parks	Number of parks		
	Density of industrial facilities	Number of factories		
	Density of transportation facilities	Sum of bus stops, subway stations, and parking lots		
Features of street users	Unweighted Street Segment betweenness centrality	Calculation of intermediary centrality of street segments		
	Local permanent resident flow weighted betweenness centrality	Extract the centerline of the street segment, then extend 55 m to both sides of the road to create a road buffer zone, generate Thiessen polygons from the base station, count the total number of mobile signaling within each polygon, and calculate the population density within each Thiessen polygon, which can be used to obtain the spatial distribution of population density. This study assumes that the population within the Thiessen polygon is uniformly distributed and deduced the population density within the street buffer as the street weights; then, the weighted betweenness centrality is calculated [15]		Mobile signaling data
	Local floating resident flow weighted betweenness centrality			
	Traveler flow weighted betweenness centrality	Extract the centerline of the street segment; then, extend 55 m to both sides of the road to create a road buffer zone, the number of points falling within the buffer zone is summed and then divided by the area of the region to obtain the density of the trajectory points as the street weights and then weighted betweenness centrality calculation		Two Steps website
Features of social structure	Percentage of immigrants	Number of people in the region without Guangzhou registered permanent residence/total number of people living in the region	Referring to Kim's (2018) study using the SA imputation method to assign census data to street segments [16]	Guangzhou 2020 Census Yearbook
	Percentage of youth population	Number of people aged 19–30 in the region/total population living in the region		

relationship between the number of occurrences of an event and the probability of that event occurring under a certain condition [19], which can be a better fit for regression relationships between crime occurrences and over-discrete distributions of different variables in complex urban street segments and their environments. In a negative binomial regression model, the dependent variable is a count variable indicating the number of times a crime event occurs. Meanwhile, the independent variables are a set of predictor variables indicating the factors that may be associated with the occurrence of the event. Hypothesis $\widehat{y_i}$ represents the frequency of crime incidents in a given street segment, which shows a negative binomial distribution. The model equation is as follows:

$$\ln(\widehat{y_i}) = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \ldots + \beta_m X_{im} + \varepsilon_i, m = 1, 2, \ldots, M, \tag{3}$$

where m denotes the m-th independent variable, β_0 is the intercept, β_m is the coefficient of the independent variable X_{im}, and ε_i denotes the residuals of the model. The negative binomial regression model increases the estimation of residual ε_i, usually by making ε_i obey the *Gamma* distribution:

$$\varepsilon \sim Gamma\left(\frac{1}{\alpha}, \alpha\right), \ \alpha > 0. \tag{4}$$

This study first calculates and assigns the betweenness centrality of street segments as features of street users and then introduces two sets of control variables, namely features of built environment and social structure. In addition, it constructs four regression models to regress the number of street crimes on the built environment, street users, and social structure features, respectively, to explore the correlation between urban street network characteristics and crimes.

4 Results and discussion

4.1 Analysis of the correlation between features of street users and crime

Table 2 shows the negative binomial regression results. From the model results, the unweighted street segment betweenness centrality and traveler flow weighted street segment betweenness centrality have a significant positive impact relationship on crime; the local floating transient flow weighted street segment betweenness centrality has a significant negative impact relationship on crime, while the local permanent resident flow weighted street segment betweenness centrality has no significant impact relationship on crime. Figure 2 shows the distribution of street segments with different types of betweenness centrality in the street network.

Unweighted street segment betweenness centrality measures the popularity or potential for use of a particular street segment, which from a criminological perspective, represents the collective space of consciousness that people (including criminals) may develop in the course of their daily activities [5]. The higher the betweenness centrality of a street segment, the higher the potential concentration of activity, the more likely that criminals and targets will congregate at the same time and place, and the higher crime risk the street segment will face. In addition, street segments with higher betweenness centrality have better passage potential; and the public will traverse, stay, and gather more frequently. The cost of crime location selection is lower, and criminals can quickly get away from their victims or pursuers by examining the escape conditions of the street and completely exploiting the high degree of connectivity of the road network. Consequently, high-crime areas tend to have highly trafficked roadways passing through them.

The local floating population comprises the people who flow into the local area from other places and live in the local area. Its weighted betweenness centrality measures the use potential and potential monitoring degree of the floating population on the street segment, which has a significant negative impact on crime and somewhat indicates that the surveillance of the local floating resident flow has an inhibitory effect on crime. Areas near streets with higher betweenness centrality of local floating resident flow, where the intensity of activity of the local population is high and the distinction between public and private space is high, add visibility to strangers and vigilance among residents. This scenario facilitates the provision of informal surveillance by locals living on or near the street and reduces the risk of crime.

The betweenness centrality of traveler weighted street segments has a significant positive relationship on crime. On the one hand, the anonymity of streets with high travel crowds increases, ambiguity arises as to who is responsible for the space, strangers provide less informal supervision, and non-local activities inhibit the ability of local residents to show territoriality and deter crime, which can easily lead to crime. On the other hand, travel hotspots tend to have mixed crowds, which causes difficulty in supervising and managing security, making it easier for criminals to hide in the crowd and escape after committing crimes and providing favorable conditions for criminals to choose their targets and carry out criminal activities.

4.2 Analysis of the correlation between features of built environment, social structure, and crime

The model regression results generally show that the built environment feature factors of the densities of commercial facilities, business facilities, recreational facilities, medical and health facilities, industrial facilities, and transportation facilities have a significant positive impact on crime. Park density has a significant positive impact on the number of crimes in regression models 2–4. Meanwhile, the density of cultural facilities, educational and scientific research facilities, and residential and residential service facilities had no impact on crime. The percentage of immigrants has a significant negative effect on the number of crimes in the social structure feature factor, whereas the percentage of youth population lacks a significant effect on crime.

The positive impact of commercial facilities on crime is consistent with the findings of previous research. Numerous studies have shown that many types of commercial facilities are desirable for gathering people and facilitating the escape of crime [20]. Among them, small stores, restaurants and convenience stores, supermarkets, food markets, and newsstands have a strong proximity to the street, are more frequently used by residents in their daily lives, and are more attractive to criminal activities. Parts of the hotel, auto repair stores, building materials, and furniture stores around the specific time have more people. However, the necessary management is lacking with weak surveillance, thereby also easily

attracting crimes. Bars, as places where people mix, are prone to gather potential criminals and criminal targets. Bars are also prone to gather people who are impulsive to commit crimes due to alcohol abuse. Additionally, drunken people may be less aware of their precautions, which makes them more likely to become criminal targets and attract criminals to commit crimes.

The business facilities in this study include banks and companies, which have a significant positive impact on crime. Banks are usually located along streets with high levels of crowd activity, have higher pedestrian flow and utilization rates, closely associate with wealth aggregation, and with wealth exposure to users, making them attractive targets to property crime [14]. Companies have weaker connections and interactions with the street. Thus, street surveillance plays a relatively weak role for companies, making them prone to become crime aggregators.

The recreational facilities have a significant positive influence relationship on crime. These facilities often act as crime attractors, which can largely provide criminals with opportunities to commit crimes and become typical locations for repeat crimes with specific motives [21]. On the one hand, the high degree of crowd mixing in recreational venues is favorable for criminals to hide and become places for criminals to case and observe. On the other hand, the crowd inside the venues stay for a long time, which prevent them from playing an effective role in monitoring the street activities. They are also not conducive to creating good order on the street, making it easily susceptible to become a crime-sensitive place.

Medical and health facilities have a significant positive impact on crime. Medical and health facilities are closely linked to residents' daily activities, have a higher mix of people entering and exiting them, and have a higher proportion of vulnerable groups, thus increasing their likelihood to become potential crime targets. The population may also be less vigilant against crime by focusing on their own health, which reduces the role of surveillance of the surrounding environment and makes the vicinity of such places attractive to crime.

Compared with other facilities, factories have greater spatial independence and weaker interaction with the streets, resulting in fewer street crime targets. However, their natural surveillance of the streets is also weaker. At night and other times of the day, they may form a surveillance dead zone, which is conducive to the implementation of criminal activities. In addition, most of the factories are manufacturing factories, in which several laborers belong to low-income groups with low education levels and are prone to commit crimes against people in the surrounding areas when they are subjected to undesirable inducements or long hours of labor that lead to greater psychological pressure [22].

Transportation facilities had a significant positive relationship with crime. Moreover, their regression coefficients were the largest of all significant variables in all four models. Transportation facilities are often adjacent to roads, closely related to street activities, and carry a large number of complex flows of people, making it easy for criminals to hide in them, providing other opportunities for crime and a stronger attraction to commit crime. Transportation facilities are generally located at convenient intersections and streets, providing favorable conditions for criminals to achieve rapid escape and easily become crime hot spots.

Park density produces a significant positive relationship on crime in Regression Models 2, 3, and 4, as reflected in the promotion of criminal activity. Related studies have shown that the level of surveillance is an important factor in explaining the relationship between parks and crime [23]. Parks, as highly open places, attract potential criminals. As public places with no clear ownership, parks have a weaker role of mutual surveillance by strangers, which can easily create opportunities for criminals. Meanwhile, some of the parks have problems such as overgrowth of vegetation, insufficient lighting, and low security guards, which likewise weaken the level of surveillance and are prone to robberies and aggressions.

In terms of street social structure features, the percentage of immigrants in this study produced a significant negative relationship on crime, which is different from the findings of previous studies [4]. The higher instability of undocumented, provisional, and refuge migrants may dilute the social ties of local residents, which is not conducive to informal control. Owing to the influence of relevant policies and the attractiveness of big cities, Guangzhou has a high proportion of talented and practicing foreign employees who are relatively stable in their daily activities and can form informal control over the streets and have the ability to protect criminal targets through surveillance.

5 Conclusions

Taking the four central districts of Guangzhou as an example, this study constructs a complex network of urban street space and carries out a negative binomial correlation analysis between street crime and the features of built environment, street users, and social structure. The aim is to excavate the characteristics of the urban street environment that are closely related to the criminal activities and to put forward the relevant suggestions for shaping the safety-targeted urban street space and constructing a safe city.

One of the main contributions of this study is to characterize the passage potential of streets and different types of crowd surveillance using a street betweenness centrality indicator weighted by pedestrian flow data, completing the research that has used the concept of street betweenness to reveal the impact of street network environments on crime. The results show that streets with higher unweighted betweenness centrality and traveler weighted betweenness centrality are prone to crime. In addition, these streets have better passage potential and stronger anonymity with a lower degree of informal surveillance, which makes it easy for criminals to hide and escape and thus should be subjected to enhanced control. Moreover, mechanical and formal surveillance should be supplemented in areas where natural surveillance is insufficient. The weighted betweenness centrality of local floating resident flow has a negative effect on crime. Streets with a large number of locals are conducive to the provision of informal guardianship by locals living on or near the street, which helps curb crime. Local construction should be strengthened to form a favorable regional atmosphere and give full play to the surveillance provided by the local floating population on the streets.

Another contribution of this study is to construct a comprehensive spatial pattern of crime by considering the influence of the built environment and social structure features of streets. The results show that the density of commercial, business, recreational,

Table 2: Negative Binomial Regression Model Results.

Form	Indicator	Regression coefficient			
		Model 1	Model 2	Model 3	Model 4
Features of the built environment	Density of commercial facilities	6.233**	6.249**	6.270**	6.205**
	Density of business facilities	1.613**	1.579**	1.517**	1.629**
	Density of recreational facilities	1.114**	1.092**	1.057**	1.067**
	Density of cultural facilities	0.174	0.156	0.155	0.139
	Density of educational and scientific research facilities	−0.931	−0.990	−1.100	−1.013
	Density of medical and health facilities	1.233**	1.196**	1.167**	1.240**
	Density of residential and residential service facilities	0.054	0.027	0.000	0.036
	Density of Parks	0.503	0.516*	0.551*	0.515*
	Density of industrial facilities	2.508**	2.528**	2.520**	2.493**
	Density of transportation facilities	7.530**	7.764**	7.932**	7.675**
Features of street users	Unweighted Street Segment betweenness centrality	1.001*			
	Local permanent resident flow weighted betweenness centrality		0.183		
	Local floating resident flow weighted betweenness centrality			−4.507*	
	Traveler flow weighted betweenness centrality				1.557**
Features of social structure	Percentage of immigrants	−1.880**	−1.859**	−1.829**	−1.830**
	Percentage of youth population	0.112	0.081	0.019	0.105

Note: *, ** and *** denote significant at 10%, 5%, and 1% confidence levels, respectively.

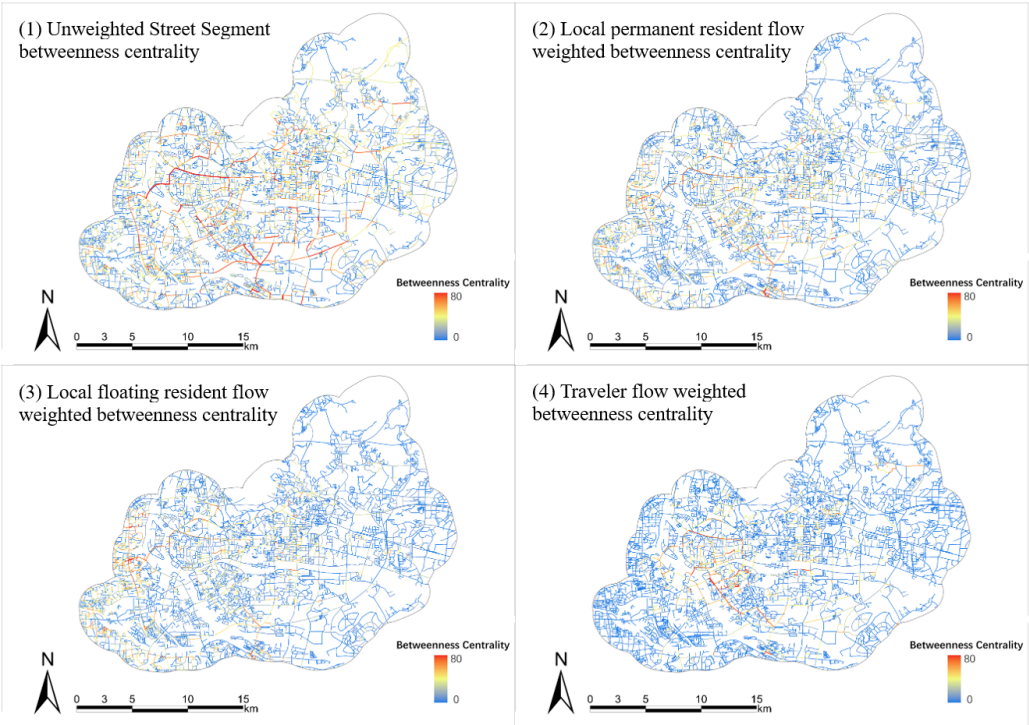

Figure 2: Distribution of streets with different types of betweenness centrality in the street network.

medical and health, industrial, and transportation facilities near the streets positively affects criminal activities, and that these locations, where daily pedestrian flows are dense and mixed, should be targeted with appropriate regulatory measures according to the types and characteristics of the facilities to reduce the threat of crime. In terms of social structure features, the percentage of immigrants in this study has a negative impact on crime, strengthening the need for urban security to pay attention to the mobility of the population, focus on the formation of social cohesion among residents, ensure the stability of street activities, and strengthen the joint defense and

joint governance to maintain street order. This study can also help in proposing appropriate planning and policy recommendations for the refined management and engineering of urban.

The current study contains some limitations. On the one hand, street crime cases are categorized into robberies, thefts, and other types; and the impact mechanisms of different types of criminal activities may differ. Thus, future research can further subdivide the types of cases to draw more targeted conclusions. On the other hand, the impact of street environment factors on crime may have temporal differences, and the current model cannot explain the temporal effects of the factors, which may vary at different times of the day or days of the week. In the future, when constructing the street complex network model, we can consider incorporating elements such as street type, speed limit, and barriers between the starting point and destination to describe the complex street spatial environment more comprehensively and perfectly and to excavate deeper mechanisms of the role of street criminal activities.

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